

Recognize, Don't Generate: Anonymous Supplementary Material

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1 Scope and reporting policy

This supplement records the protocols used for the representation and temporal QA diagnostics. We report complete valid tasks regardless of the sign of the pretrained-random difference. A linear probe measures accessibility under its readout; it is not treated as evidence that the model understands the property or uses it in forecasting or question answering. Native zero-shot QA and diagnostics with a trained head are reported as different interfaces.

2 Property-specific representation probes

The Level-1 study contains ten model checkpoints, pretrained and same-architecture random initializations, and seeds 7, 17, and 27. Each of the 60 frozen-representation caches is reused across property probes. Samples, splits, readout class, and preprocessing are matched within every pretrained-random pair. We aggregate paired differences after orienting every metric so that a positive value means that pretraining is better. Thus the difference for balanced accuracy and R^2 is pretrained minus random, whereas the difference for NMAE is random minus pretrained.

Property	Metric	Pretrained	Random	Improvement	Positive models
Dominant period	balanced acc.	0.715	0.364	+0.351	10/10
Anomaly present	balanced acc.	0.754	0.599	+0.155	10/10
Local dependence	balanced acc.	0.460	0.331	+0.129	10/10
Noise level	balanced acc.	0.542	0.436	+0.107	10/10
Dominant period	NMAE ↓	0.200	0.594	+0.394	10/10
Local parameter	NMAE ↓	0.647	0.861	+0.213	10/10
Trend strength	NMAE ↓	0.872	0.885	+0.013	6/10

Table 1: Final-layer means across ten models and three seeds. The mixed trend result is retained rather than filtered by its sign.

3 Relation and composition protocol

Every Level-2 run consumes a seed-specific manifest shared by all models and initializations. Manifest seeds are derived with SHA-256 rather than the process-randomized Python hash. The manifest stores the representation-pool hash, sample indices, train/test split, task labels, and protocol hash. This makes the pair and split identity checkable across processes.

The audit changed three task definitions. First, the unseen-combination C1 test has constant trend and local labels, so balanced accuracy is reported only for label axes with test support for at least two classes. Second, context-only standardization removes the scale information required by the earlier volatility-ordering R3 task; R3 is omitted with that reason instead of being reported as a valid negative result. Third, the earlier R7 duplicated R4. R7 is now a three-series ordering problem: given an anchor and two candidates, the label identifies which candidate is closer in the relevant latent parameter.

Corrected results are written to a versioned directory and are never merged with protocol-v1 outputs. We make no ten-model Level-2 claim until paired coverage is complete.

4 IRTS temporal-relation protocol

For each of 150 temporal_relationship items, the adapter extracts the prefix values, prefix timestamps, and full timestamps. The question branch contains the question and answer options after removing all labelled temporal arrays. The temporal branch contains only the labelled values and timestamps. Question features always come from the pretrained cache in the factorized diagnostic; the pretrained and random comparison changes only the temporal branch. Item IDs, features, task types, and gold answers are filtered by one shared mask. A single linear head is fitted on the combined branch training features and evaluated separately for the two temporal initializations.

Shuffling and reversal operate on the original numeric lexemes. They preserve sign, exponent, and decimal precision exactly, while timestamps remain unchanged. Answer parsing follows the released evaluator implementation.

The legacy native runner right-padded decoder-only batches, so shorter prompts generated after padding tokens. We withhold all legacy native scores. Its question-only, shuffled, and reversed contrasts also used incomplete temporal extraction and inconsistent numeric precision. The corrected native runner uses left padding, exact numeric lexemes, and all labelled value and timestamp arrays. For official Qwen3-8B-Base, accuracy/parse coverage is 0.153/0.473 (full), 0.193/0.947 (question-only), 0.167/0.473 (shuffled), and 0.133/0.473 (reversed). The unequal parse rates prevent a direct language-prior or temporal-use interpretation of the accuracy differences.

For the first corrected paired caches, the DeepSeek-V2-Lite shared-head diagnostic obtains balanced accuracy 0.292 with the pretrained temporal branch and 0.294 with the random branch in the full condition; official Qwen3-8B-Base obtains 0.348 and 0.322. Question-only is branch-identical by construction (0.371 and 0.390, respectively). Shuffle and reversal do not yield a consistent order effect; Qwen’s shuffled pretrained score (0.390) is above its full score. These two single-seed results do not show a consistent pretrained temporal-branch advantage. This boundary prevents a high probe score on synthetic data from being presented as evidence of strong IRTS QA.

5 Artifact provenance

The Level-1 table is generated from the 60 per-run CSV shards rather than a manually transcribed table. Corrected Level-1 aggregates, Level-2 manifests, and QA outputs are stored beneath a versioned protocol directory. The paper’s claim-evidence matrix records, for each quantitative statement, the raw path, metric, model scope, seed scope, and protocol version. Legacy outputs remain available for audit but are excluded from corrected aggregates.